THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** IC	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
2	**(d)** 73	Multiply each digit by the power of the base and add
3	**(c)** 322	Divide repeatedly by the base and read the remainders upwards
4	**(d)** 135	Base 60: value = 60 x first + second
5	**(d)** 72	Base 20: value = 20 x first + second
6	**(a)** Hindu-Arabic system	Digits 0-9 with place value came from India and spread through Arabs
7	**(d)** VX	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
8	**(d)** 125	Multiply each digit by the power of the base and add
9	**(c)** 226	Divide repeatedly by the base and read the remainders upwards
10	**(b)** 220	Base 60: value = 60 x first + second

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3 \times 5^2 + 1 \times 5^1 + 4 \times 5^0$ (1); $= 84$ (1).	1+1
12	$29 = 2 \times 14 + 1$; $14 = 2 \times 7 + 0$; $7 = 2 \times 3 + 1$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (1); 11101_2 (1).	1+1
13	Babylonian: $3 \times 60 + 25 = 205$ (1); Maya: $4 \times 20 + 11 = 91$ (1).	1+1
14	$2 \times 8^2 + 6 \times 8^1 + 3 \times 8^0$ (1); $= 179$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$137 = 5 \times 27 + 2$; $27 = 5 \times 5 + 2$; $5 = 5 \times 1 + 0$; $1 = 5 \times 0 + 1$ (1); 1022_5 (1); check $1 \times 5^3 + 0 \times 5^2 + 2 \times 5^1 + 2 \times 5^0 = 137$ (1).	1+1+1
16	XLIV = 44, LXVII = 67 (1); sum = 111 (1); CXI (1).	1+1+1
17	$2 \times 5^3 + 4 \times 5^2 + 3 \times 5^1 + 1 \times 5^0$ (2); $= 366$ (1).	2+1
	OR: $214 = 8 \times 26 + 6$; $26 = 8 \times 3 + 2$; $3 = 8 \times 0 + 3$ (1); 326_8 (1); check $3 \times 8^2 + 2 \times 8^1 + 6 \times 8^0 = 214$ (1).	1+1+1
18	LXXXIX = 89, CXXIII = 123 (1); sum = 212 (1); CCXII (1).	1+1+1
	OR: $3 \times 8^3 + 7 \times 8^2 + 0 \times 8^1 + 4 \times 8^0$ (2); $= 1988$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Column addition with carries gives 10100_2 (3); decimal check: $13 + 7 = 20$ and $10100_2 = 20$ (2).	3+2
	OR: $13 = 1101_2$ (2); weights used: 1, 4, 8 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1
20	Column addition with carries gives 100111_2 (3); decimal check: $25 + 14 = 39$ and $100111_2 = 39$ (2).	3+2
	OR: $21 = 10101_2$ (2); weights used: 1, 4, 16 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1

Mistakes to avoid:

- Repeating V, L or D, or writing IC, IL, XM for 99, 49, 990; use XCIX, XLIX and CMXC.
- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 107	Base 20: value = 20 x first + second
2	**(c)** V, L and D	Only I, X, C and M may be repeated (up to three times)
3	**(a)** IL	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
4	**(a)** 46	Multiply each digit by the power of the base and add
5	**(a)** 1212	Divide repeatedly by the base and read the remainders upwards
6	**(b)** 85	Base 60: value = 60 x first + second
7	**(c)** 224	Base 20: value = 20 x first + second
8	**(b)** 3 times	III is 3; IIII is not used
9	**(a)** XM	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
10	**(a)** 201	Multiply each digit by the power of the base and add

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$37 = 2 \times 18 + 1$; $18 = 2 \times 9 + 0$; $9 = 2 \times 4 + 1$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (1); 100101_2 (1).	1+1
12	Babylonian: $2 \times 60 + 48 = 168$ (1); Maya: $6 \times 20 + 9 = 129$ (1).	1+1
13	$2 \times 3^3 + 1 \times 3^2 + 0 \times 3^1 + 2 \times 3^0$ (1); $= 65$ (1).	1+1
14	$45 = 2 \times 22 + 1$; $22 = 2 \times 11 + 0$; $11 = 2 \times 5 + 1$; $5 = 2 \times 2 + 1$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (1); 101101_2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$89 = 3 \times 29 + 2$; $29 = 3 \times 9 + 2$; $9 = 3 \times 3 + 0$; $3 = 3 \times 1 + 0$; $1 = 3 \times 0 + 1$ (1); 10022_3 (1); check $1 \times 3^4 + 0 \times 3^3 + 0 \times 3^2 + 2 \times 3^1 + 2 \times 3^0 = 89$ (1).	1+1+1
16	CLVI = 156, CCXLIX = 249 (1); sum = 405 (1); CDV (1).	1+1+1
17	$2 \times 3^4 + 1 \times 3^3 + 0 \times 3^2 + 1 \times 3^1 + 2 \times 3^0$ (2); $= 194$ (1).	2+1
	OR: $175 = 6 \times 29 + 1$; $29 = 6 \times 4 + 5$; $4 = 6 \times 0 + 4$ (1); 451_6 (1); check $4 \times 6^2 + 5 \times 6^1 + 1 \times 6^0 = 175$ (1).	1+1+1
18	CCLXXV = 275, CCCXVIII = 318 (1); sum = 593 (1); DXCIII (1).	1+1+1
	OR: $5 \times 6^3 + 0 \times 6^2 + 4 \times 6^1 + 3 \times 6^0$ (2); $= 1107$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Column addition with carries gives 11110_2 (3); decimal check: $19 + 11 = 30$ and $11110_2 = 30$ (2).	3+2
	OR: $27 = 11011_2$ (2); weights used: 1, 2, 8, 16 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1
20	Column addition with carries gives 110011_2 (3); decimal check: $30 + 21 = 51$ and $110011_2 = 51$ (2).	3+2
	OR: $35 = 100011_2$ (2); weights used: 1, 2, 32 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1

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- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1210	Divide repeatedly by the base and read the remainders upwards
2	**(c)** 245	Base 60: value = 60 x first + second
3	**(c)** 289	Base 20: value = 20 x first + second
4	**(a)** zero and negative numbers	Brahmagupta described how to add, subtract and multiply with zero and negatives
5	**(b)** VL	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
6	**(d)** 106	Multiply each digit by the power of the base and add
7	**(d)** 235	Divide repeatedly by the base and read the remainders upwards
8	**(d)** 330	Base 60: value = 60 x first + second
9	**(c)** 58	Base 20: value = 20 x first + second
10	**(d)** 20	Maya numerals used dots, bars and a shell with base 20

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Babylonian: $5 \times 60 + 12 = 312$ (1); Maya: $8 \times 20 + 14 = 174$ (1).	1+1
12	$1 \times 6^2 + 4 \times 6^1 + 5 \times 6^0$ (1); $= 65$ (1).	1+1
13	$53 = 2 \times 26 + 1$; $26 = 2 \times 13 + 0$; $13 = 2 \times 6 + 1$; $6 = 2 \times 3 + 0$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (1); 110101_2 (1).	1+1
14	Babylonian: $4 \times 60 + 33 = 273$ (1); Maya: $9 \times 20 + 3 = 183$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$250 = 4 \times 62 + 2$; $62 = 4 \times 15 + 2$; $15 = 4 \times 3 + 3$; $3 = 4 \times 0 + 3$ (1); 3322_4 (1); check $3 \times 4^3 + 3 \times 4^2 + 2 \times 4^1 + 2 \times 4^0 = 250$ (1).	1+1+1
16	XLIX = 49, CMLI = 951 (1); sum = 1000 (1); M (1).	1+1+1
17	$3 \times 4^4 + 0 \times 4^3 + 2 \times 4^2 + 1 \times 4^1 + 3 \times 4^0$ (2); $= 807$ (1).	2+1
	OR: $333 = 7 \times 47 + 4$; $47 = 7 \times 6 + 5$; $6 = 7 \times 0 + 6$ (1); 654_7 (1); check $6 \times 7^2 + 5 \times 7^1 + 4 \times 7^0 = 333$ (1).	1+1+1
18	CCCLXII = 362, CDXXXVIII = 438 (1); sum = 800 (1); DCCC (1).	1+1+1
	OR: $6 \times 7^3 + 1 \times 7^2 + 5 \times 7^1 + 0 \times 7^0$ (2); $= 2142$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Column addition with carries gives 100100_2 (3); decimal check: $27 + 9 = 36$ and $100100_2 = 36$ (2).	3+2
	OR: $44 = 101100_2$ (2); weights used: 4, 8, 32 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1
20	Column addition with carries gives 101000_2 (3); decimal check: $22 + 18 = 40$ and $101000_2 = 40$ (2).	3+2
	OR: $51 = 110011_2$ (2); weights used: 1, 2, 16, 32 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1

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